Quantum Teleportation with Python and Qiskit

A complete study guide: entanglement, Bell states, the full mathematical derivation,
circuit diagrams, tested code, verification experiments and viva questions

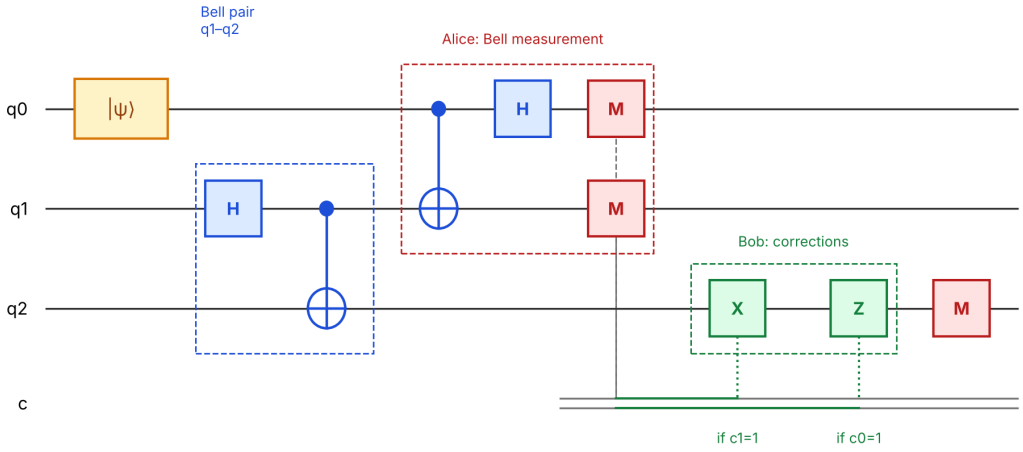


Figure 1: The teleportation circuit. q0 holds the message |ψ⟩; q1 and q2 form the shared Bell pair.

Python • Qiskit • Qiskit Aer • Entanglement • Bell measurement • Classical feed-forward

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Important
Two corrections to the notes you shared. (1) The sample output listed there shows only four outcomes (000, 011, 101, 110). I ran the code and this circuit produces **eight** outcomes, each with probability 1/8 (about 125 of 1000 shots). The reason is explained in Section 7. (2) Measuring Bob's qubit in the Z basis after teleporting |+⟩ only shows a 50/50 split, which would also happen if teleportation had failed. Section 10 gives a test that actually proves it worked.

1. Aim

To implement the quantum teleportation protocol using Python and Qiskit, transferring the state of one qubit (Alice's message qubit) onto a distant qubit (Bob's qubit) using a shared entangled pair and two classical bits, and to verify that the transfer succeeded.

2. What is Quantum Teleportation?

Definition
Quantum teleportation is a protocol that transfers an unknown quantum state from one qubit (Alice's) to another qubit (Bob's), using one pair of entangled qubits and two bits of ordinary classical communication. No qubit travels from Alice to Bob; the *state* is re-created on Bob's qubit while the original is destroyed.

2.1 The problem it solves

Alice holds a qubit in an unknown state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. She wants Bob, who is far away, to end up with a qubit in the same state. There are obstacles:

- **She cannot just measure it and tell Bob the result.** One measurement gives a single bit and destroys the state. The two numbers α and β (complex, continuous) cannot be recovered from one measurement.
- **She cannot copy it.** The no-cloning theorem forbids copying an unknown quantum state.
- **She could physically send the qubit,** but the channel may be noisy or impossible (for example, a fragile qubit over a long distance).

Teleportation sidesteps all three: the heavy lifting is done by entanglement shared *beforehand*, and afterwards only two ordinary bits need to be sent.

2.2 What teleportation is NOT

Myth	Reality
It moves matter or energy	No. Only the quantum state (information) is re-created on Bob's qubit. Bob's qubit was already there.
It is faster than light	No. Bob cannot finish until Alice's two classical bits arrive, and those travel at or below light speed.
It copies the state	No. Alice's measurement destroys her original. At the end there is exactly one copy, on Bob's side.
Bob can read the state immediately	No. Before the correction, Bob's qubit is in one of four possible states and looks completely random to him.

2.3 What is consumed

Resource	Amount per teleported qubit
Entangled pair (Bell state)	1
Classical bits sent Alice → Bob	2
Qubits Alice must hold	2 (message + her half of the pair)

3. Basic Terminology in Depth

3.1 Quantum state and ket notation

A single-qubit state is $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$. The symbol $|\psi\rangle$ is a **ket** from Dirac notation: it names a state vector. $|0\rangle = (1, 0)^T$ and $|1\rangle = (0, 1)^T$ are the computational basis states, and $|0\rangle$ is the name of a vector, not the number 0. Measuring gives 0 with probability $|\alpha|^2$ and 1 with probability $|\beta|^2$ (Born rule). Two-qubit labels such as $|01\rangle$ mean $|0\rangle \otimes |1\rangle$.

3.2 Tensor product

To describe several qubits together we combine their vectors with the tensor product: $(a, b)^T \otimes (c, d)^T = (ac, ad, bc, bd)^T$. Three qubits need $2^3 = 8$ amplitudes. In this experiment the full state starts as $|\psi\rangle \otimes |0\rangle \otimes |0\rangle$ and we track it through each gate.

3.3 Entanglement

Two qubits are **entangled** if their joint state cannot be written as a product of separate states. The classic example is

$$|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$$

Why it cannot be a product. Suppose it equalled $(a|0\rangle+b|1\rangle)\otimes(c|0\rangle+d|1\rangle) = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle$. Matching coefficients needs $ad = 0$ and $bc = 0$, yet $ac \neq 0$ and $bd \neq 0$. Impossible, so the pair is entangled.

What entanglement does and does not give. Each qubit alone, looked at in the Z basis, is a perfectly random bit. But the two results always *agree*. This correlation is the resource that teleportation uses. It does **not** let Alice send a message by measuring her half, because Bob's results look random until he is told something.

3.4 Bell states

Name	State	Meaning of the label
$\|\Phi^+\rangle$	$(\|00\rangle + \|11\rangle)/\sqrt{2}$	Equal, same bits, plus sign
$\|\Phi^-\rangle$	$(\|00\rangle - \|11\rangle)/\sqrt{2}$	Same bits, minus sign
$\|\Psi^+\rangle$	$(\|01\rangle + \|10\rangle)/\sqrt{2}$	Opposite bits, plus sign
$\|\Psi^-\rangle$	$(\|01\rangle - \|10\rangle)/\sqrt{2}$	Opposite bits, minus sign (the singlet)

The four Bell states are mutually orthogonal and together form a **basis** for two qubits, called the **Bell basis**. Any two-qubit state can be written as a mixture of these four. Teleportation relies on measuring Alice's two qubits *in the Bell basis*.

3.5 How to create $|\Phi^+\rangle$

1. Start with $|00\rangle$.
2. Apply H to the first qubit: $(|0\rangle + |1\rangle)/\sqrt{2} \otimes |0\rangle = (|00\rangle + |10\rangle)/\sqrt{2}$.
3. Apply CNOT (control = first, target = second): $|00\rangle \to |00\rangle$, $|10\rangle \to |11\rangle$. Result: $(|00\rangle + |11\rangle)/\sqrt{2} = |\Phi^+\rangle$.

In our circuit the Hadamard is applied to q1 and CNOT has control q1 and target q2, so the Bell pair lives on (q1, q2).

3.6 Quantum gates used

$$H = \frac{1}{\sqrt{2}}\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}$$ creates superposition: $H|0\rangle = (|0\rangle+|1\rangle)/\sqrt{2}$

$$X = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}$$ bit flip: $X|0\rangle = |1\rangle$, $X|1\rangle = |0\rangle$

$$Z = \begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix}$$ phase flip: $Z|0\rangle = |0\rangle$, $Z|1\rangle = -|1\rangle$

CNOT (controlled-NOT) acts on two qubits and flips the target only when the control is $|1\rangle$:

$$\text{CNOT} = \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{vmatrix}$$

Input $\|$control target$\rangle$	Output
$\|00\rangle$	$\|00\rangle$
$\|01\rangle$	$\|01\rangle$
$\|10\rangle$	$\|11\rangle$
$\|11\rangle$	$\|10\rangle$

All of these gates are **unitary** (reversible). The only irreversible step in the protocol is measurement.

3.7 Measurement and classical bits

Measuring a qubit gives 0 or 1 and collapses its state. The result is stored in a **classical bit**. In our circuit, c0 holds the result of measuring q0 and c1 holds the result for q1. These two bits are the message Alice sends to Bob.

3.8 Classical feed-forward

Feed-forward means using a measurement result *during* the circuit to decide whether to apply a later gate. In Qiskit this is written with `with qc.if_test((bit, value)):`. Here it models Bob receiving Alice's two bits and applying X and/or Z accordingly.

3.9 No-cloning theorem

It is impossible to build a machine that copies an arbitrary unknown quantum state. Teleportation is consistent with this: the original is destroyed by Alice's measurement before Bob's copy is complete.

3.10 LOCC

Teleportation uses only **Local Operations and Classical Communication** plus pre-shared entanglement. Alice and Bob each act only on their own qubits and talk over a classical channel. That is why teleportation is a statement about entanglement as a resource.

4. Why Teleportation Works: the Big Picture

4.1 Roles

Qubit	Owner	Role
q0	Alice	Holds the unknown state $\|\psi\rangle$ (the message)
q1	Alice	Her half of the Bell pair
q2	Bob	His half of the Bell pair; receives the state

4.2 Intuition

Imagine Alice and Bob previously met, created an entangled pair, and each took one qubit home. Later, Alice wants to send $|\psi\rangle$. She performs a joint measurement on $|\psi\rangle$ and her half of the pair. This joint measurement does two things at once:

1. It **destroys** the state on her side, so no cloning occurs.
2. It **forces Bob's qubit** into one of four states, each of which is $|\psi\rangle$ with a simple error (nothing, X, Z, or both). Which error occurred is exactly what her two measured bits say.

Alice's two bits are completely random (each of four values has probability 1/4) and are independent of $|\psi\rangle$. So someone intercepting the classical bits learns nothing about α and β. Bob uses the bits to undo the error and recovers $|\psi\rangle$.

4.3 Analogy (and its limits)

Think of Bob's qubit as a locked box and Alice's two bits as which of four keys opens it. The contents ($|\psi\rangle$) were put in the box instantly by entanglement and measurement, but the key must travel the ordinary way. The analogy is imperfect (quantum states are not hidden values), but it captures why classical communication is essential.

5. The Full Mathematical Derivation

We write each three-qubit basis state as |q0 q1 q2⟩. (Note: Qiskit *prints* results in the opposite order, c2 c1 c0. Here we use the physics order q0 q1 q2.)

Step 0: Initial state

$$|\Psi_0\rangle = |\psi\rangle \otimes |0\rangle \otimes |0\rangle = (\alpha|0\rangle + \beta|1\rangle) \otimes |00\rangle$$

Step 1: Create the Bell pair on q1, q2

$$|\Psi_1\rangle = |\psi\rangle \otimes (|00\rangle + |11\rangle)/\sqrt{2} = (1/\sqrt{2})[\ \alpha|0\rangle(|00\rangle+|11\rangle) + \beta|1\rangle(|00\rangle+|11\rangle)\ ]$$

Expanded: $(1/\sqrt{2})[\ \alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle\ ]$.

Step 2: CNOT with control q0, target q1

Flip q1 whenever q0 = 1. The terms $\beta|100\rangle \to \beta|110\rangle$ and $\beta|111\rangle \to \beta|101\rangle$:

$$|\Psi_2\rangle = (1/\sqrt{2})[\ \alpha|000\rangle + \alpha|011\rangle + \beta|110\rangle + \beta|101\rangle\ ]$$

Step 3: Hadamard on q0

Use $H|0\rangle = (|0\rangle+|1\rangle)/\sqrt{2}$ and $H|1\rangle = (|0\rangle-|1\rangle)/\sqrt{2}$ on the first qubit of every term:

$$|\Psi_3\rangle = \tfrac{1}{2}[\ \alpha(|0\rangle+|1\rangle)|00\rangle + \alpha(|0\rangle+|1\rangle)|11\rangle + \beta(|0\rangle-|1\rangle)|10\rangle + \beta(|0\rangle-|1\rangle)|01\rangle\ ]$$

Step 4: Regroup by Alice's two qubits (q0 q1)

Collect all terms having the same q0 q1 value:

$$|\Psi_3\rangle = \tfrac{1}{2}[\ |00\rangle(\alpha|0\rangle+\beta|1\rangle) + |01\rangle(\alpha|1\rangle+\beta|0\rangle) + |10\rangle(\alpha|0\rangle-\beta|1\rangle) + |11\rangle(\alpha|1\rangle-\beta|0\rangle)\ ]$$

Check one group. For q0 q1 = 01 we need the terms with q0 = 0 and q1 = 1. From $\alpha(|0\rangle+|1\rangle)|11\rangle$ (q1 = 1, q2 = 1) the $|0\rangle$ part gives $|01\rangle \otimes \alpha|1\rangle$. From $\beta(|0\rangle-|1\rangle)|10\rangle$ (q1 = 1, q2 = 0) the $|0\rangle$ part gives $|01\rangle \otimes \beta|0\rangle$. Together: $|01\rangle(\alpha|1\rangle + \beta|0\rangle)$, as claimed.

Step 5: Measure q0, q1 and read the table

Each of the four outcomes has probability $(\tfrac{1}{2})^2 = \tfrac{1}{4}$ regardless of α and β, because the four bracketed states are all normalised. After Alice's result, Bob's qubit is:

Alice's result (c0, c1)	Bob's qubit q2	Correction needed	Why
(0, 0)	$\alpha\|0\rangle + \beta\|1\rangle = \|\psi\rangle$	None	Already correct
(0, 1)	$\alpha\|1\rangle + \beta\|0\rangle = X\|\psi\rangle$	X	X·X = I
(1, 0)	$\alpha\|0\rangle - \beta\|1\rangle = Z\|\psi\rangle$	Z	Z·Z = I
(1, 1)	$\alpha\|1\rangle - \beta\|0\rangle = ZX\|\psi\rangle$ (up to sign)	X first, then Z	Applying X then Z gives exactly $\|\psi\rangle$ (checked below)

Check the last row. Apply X to $\alpha|1\rangle - \beta|0\rangle$: we get $\alpha|0\rangle - \beta|1\rangle$. Apply Z: $\alpha|0\rangle + \beta|1\rangle = |\psi\rangle$. ✓

Rule: apply X if c1 = 1, then apply Z if c0 = 1

This is exactly what the code does: first `if c1: x(2)`, then `if c0: z(2)`. Order matters, because X and Z anticommute ($ZX = -XZ$); using the wrong order would only add an unobservable global sign here, but following X-then-Z is the standard rule.

> **Important**
> In the table, c0 is the result of measuring q0 and c1 is the result for q1. Do not confuse this with Qiskit's printed key order, which lists c2 c1 c0.

6. The Circuit, Step by Step

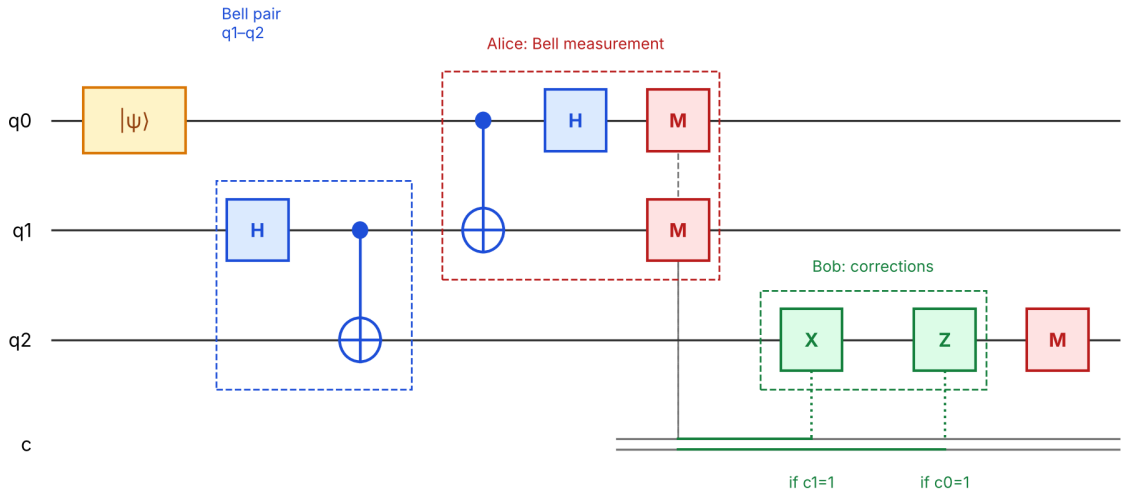


Figure 2: Teleportation circuit with the three phases highlighted: Bell-pair creation, Alice's Bell measurement and Bob's corrections.

Step	Gate (Qiskit)	Who	Purpose
1	qc.h(0)	Alice	Prepare the message state. (Here |+⟩; in general any state.)
2	qc.h(1)	Shared	Start building the Bell pair
3	qc.cx(1, 2)	Shared	Finish the Bell pair $\|\Phi^+\rangle$ on q1, q2
4	qc.cx(0, 1)	Alice	Begin Bell-basis measurement (CNOT part)
5	qc.h(0)	Alice	Finish Bell-basis measurement (Hadamard part)
6	qc.measure(0, 0); qc.measure(1, 1)	Alice	Get two classical bits
7	if c1: x(2)	Bob	Undo a possible bit flip
8	if c0: z(2)	Bob	Undo a possible phase flip
9	qc.measure(2, 2)	Bob	Read the final qubit (for testing)

6.1 Why CNOT then H is a “Bell measurement”

CNOT followed by a Hadamard on the control is the circuit that converts the four Bell states into the four computational states |00⟩, |01⟩, |10⟩, |11⟩. Real hardware can only measure in the computational basis, so to measure in the Bell basis we first rotate the Bell basis into the computational basis, and then measure. Steps 4 and 5 are this rotation.

Bell state	After CNOT + H	Measurement result
$\|\Phi^+\rangle$	|00⟩	(0, 0)
$\|\Phi^-\rangle$	|10⟩	(1, 0)
$\|\Psi^+\rangle$	|01⟩	(0, 1)
$\|\Psi^-\rangle$	|11⟩	(1, 1)

(Results shown as (q0, q1).)

7. Worked Example: Teleporting |+⟩

The code teleports |+⟩ = (|0⟩ + |1⟩)/√2, so α = β = 1/√2.

Alice's (c0, c1)	Bob's qubit before correction	Correction	Bob's qubit after
(0, 0)	(|0⟩ + |1⟩)/√2 = |+⟩	none	|+⟩
(0, 1)	(|1⟩ + |0⟩)/√2 = |+⟩	X (does nothing to |+⟩)	|+⟩
(1, 0)	(|0⟩ − |1⟩)/√2 = |−⟩	Z	|+⟩
(1, 1)	(|1⟩ − |0⟩)/√2 = −|−⟩	X then Z	|+⟩ (up to global sign)

7.1 Why the output has eight outcomes, not four

Alice's two bits are uniformly random: each of four combinations has probability 1/4. After correction, Bob's qubit is always |+⟩, and measuring |+⟩ in the Z basis gives 0 or 1 with probability 1/2 each. Alice's bits and Bob's result are independent, so each of the 4 × 2 = 8 strings has probability 1/4 × 1/2 = 1/8, i.e. about 125 of 1000 shots.

The sample output in your notes lists only 000, 011, 101 and 110. These are the strings with an even number of 1s, which would mean Bob's bit is correlated with Alice's bits. The ideal circuit produces no such correlation, and the real result is flat across all eight, as in the figure below.

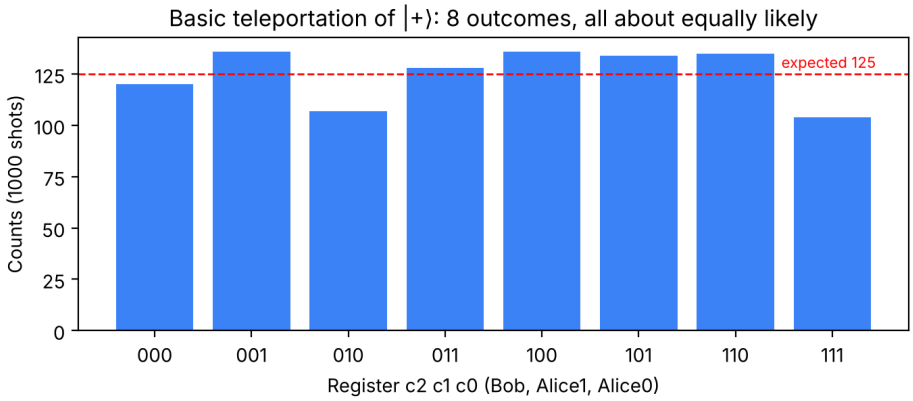


Figure 3: Result of one real run (1000 shots) of the basic code. Counts vary from run to run.

Important
Limitation of this test. |+⟩ measured in the Z basis always looks 50/50. A *failed* teleportation that left Bob's qubit in |0⟩ or |1⟩ with random outcomes would give the same histogram. So this run demonstrates the protocol, but does not prove it worked. Section 10 fixes that.

8. Code Explained Line by Line

8.1 Imports

```
from qiskit import QuantumCircuit, transpile
from qiskit_aer import AerSimulator
```

`QuantumCircuit` builds the circuit, `transpile` adapts it to the backend and `AerSimulator` runs it on your computer.

8.2 Circuit registers

```
qc = QuantumCircuit(3, 3)
```

Three qubits (q0, q1, q2) all start in |0⟩ and three classical bits (c0, c1, c2) all start at 0. We need c0 and c1 for Alice's results and c2 for Bob's final readout.

8.3 Prepare the state to teleport

```
qc.h(0)
```

H on q0 gives |+⟩ = (|0⟩+|1⟩)/√2. Replace this line with any gates to teleport a different state, for example `qc.ry(1.1, 0)`.

8.4 Create the Bell pair

```
qc.h(1)
qc.cx(1, 2)
```

`qc.cx(control, target)` is CNOT. After these two lines, q1 and q2 are in (|00⟩+|11⟩)/√2.

8.5 Alice's Bell-basis rotation

```
qc.cx(0, 1)
qc.h(0)
```

CNOT with control q0 and target q1, then H on q0. These rotate Alice's two qubits so that a computational-basis measurement is a Bell measurement (Section 6.1).

8.6 Measure Alice's qubits

```
qc.measure(0, 0)
qc.measure(1, 1)
```

Syntax: `measure(qubit, classical_bit)`. Result of q0 goes into c0 and q1 into c1.

8.7 Bob's conditional corrections

```
with qc.if_test((qc.clbits[1], 1)):
    qc.x(2)
with qc.if_test((qc.clbits[0], 1)):
    qc.z(2)
```

- `qc.clbits[1]` is the classical bit c1.
- `(qc.clbits[1], 1)` is the condition "c1 equals 1".
- Everything indented inside the `with` block runs only if the condition is true. So `qc.x(2)` runs only when Alice measured 1 on q1.
- The second block does the same for c0 and the Z gate.

- This feature is called a dynamic circuit. Older tutorials used `c_if`, which is removed in recent Qiskit versions.

8.8 Final measurement and running

```
qc.measure(2, 2)
simulator = AerSimulator()
compiled_circuit = transpile(qc, simulator)
result = simulator.run(compiled_circuit, shots=1000).result()
counts = result.get_counts()
```

Measure q2 into c2, create the simulator, transpile the circuit, run it 1000 times and obtain a dictionary of bitstrings and counts. The keys are written **c2 c1 c0** from left to right: the right-most character is classical bit 0. So '110' means c2 = 1 (Bob read 1), c1 = 1, c0 = 0.

9. Complete Code and Real Output

This is the program, exactly as I ran it:

```
# teleport_basic.py : Quantum teleportation of |+> from qubit 0 to qubit 2
# Requires: pip install qiskit qiskit-aer
from qiskit import QuantumCircuit, transpile
from qiskit_aer import AerSimulator

# 3 qubits (q0 = message, q1 = Alice's half, q2 = Bob's half) and 3 classical bits
qc = QuantumCircuit(3, 3)

# Step 1: prepare the state to teleport on q0  ->  |+> = (|0> + |1>)/sqrt(2)
qc.h(0)

# Step 2: create the Bell pair |Phi+> between q1 and q2
qc.h(1)
qc.cx(1, 2)

# Step 3: Alice's Bell-basis measurement circuit
qc.cx(0, 1)
qc.h(0)

# Step 4: Alice measures q0 -> c0 and q1 -> c1
qc.measure(0, 0)
qc.measure(1, 1)

# Step 5: Bob's corrections (classical feed-forward)
with qc.if_test((qc.clbits[1], 1)):   # if c1 == 1 apply X
    qc.x(2)
with qc.if_test((qc.clbits[0], 1)):   # if c0 == 1 apply Z
    qc.z(2)

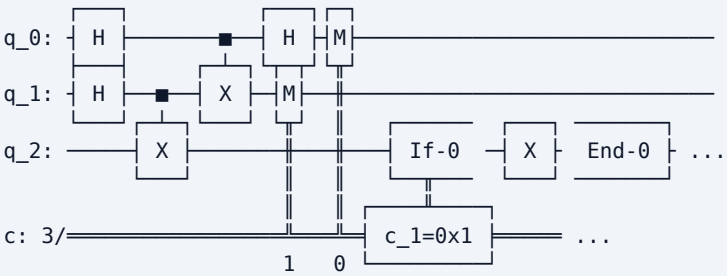
# Step 6: measure Bob's qubit into c2
qc.measure(2, 2)

print(qc.draw())

simulator = AerSimulator()
compiled_circuit = transpile(qc, simulator)
result = simulator.run(compiled_circuit, shots=1000).result()
counts = result.get_counts()

print("Teleportation Measurement Results:")
print(dict(sorted(counts.items())))
```

Circuit drawn by qc.draw()



(shortened; the real output continues with the Z block and the final measurement)

Output of one real run

```
Teleportation Measurement Results:
{'000': 120, '001': 136, '010': 107, '011': 128, '100': 136, '101': 134, '110': 135, '111': 104}
```

All eight outcomes occur with similar frequency (expected 125 each). The exact counts change every run.

> **Important**
> Requires a Qiskit/Aer version with dynamic-circuit support (`if_test`). I ran this successfully with a current installation. If you get an error about `if_test` or `c_if`, upgrade with `pip install -U qiskit qiskit-aer`.

10. A Proper Verification Test

To prove that Bob really holds $|\psi\rangle$, we use an **inverse-preparation test**:

1. Prepare an arbitrary state $|\psi\rangle = U|0\rangle$ on q0 (here U is an RY then RZ rotation).
2. Teleport it to q2.
3. Apply $U^\dagger$ (the inverse rotation) on q2.
4. If q2 holds exactly $|\psi\rangle$, then $U^\dagger|\psi\rangle = |0\rangle$, so measuring q2 gives **0 with probability 1**. If teleportation failed or corrections are missing, the probability is not 1.

We also run a **control experiment** without the correction gates. Bob's qubit is then correct only when Alice measured (0, 0), which has probability 1/4. In the other cases it carries an X error, a Z error or both, so the success rate drops to roughly 50% for this test state (the exact figure depends on the state).

```
# teleport_verify.py : a REAL test that teleportation worked
# Idea: prepare an arbitrary state, teleport it, then run the preparation
# in reverse on Bob's qubit. If Bob holds exactly that state, he gets |0>
# with probability 1.
from qiskit import QuantumCircuit, transpile
from qiskit_aer import AerSimulator

sim = AerSimulator()

def teleport_test(theta, phi, apply_corrections=True, shots=2000):
    qc = QuantumCircuit(3, 3)
    qc.ry(theta, 0)                  # prepare an arbitrary state on q0
    qc.rz(phi, 0)
    qc.h(1); qc.cx(1, 2)             # Bell pair
    qc.cx(0, 1); qc.h(0)             # Alice's Bell-basis rotation
    qc.measure(0, 0); qc.measure(1, 1)
    if apply_corrections:
        with qc.if_test((qc.clbits[1], 1)):
            qc.x(2)
        with qc.if_test((qc.clbits[0], 1)):
            qc.z(2)
    qc.rz(-phi, 2)                   # undo the preparation on Bob's qubit
    qc.ry(-theta, 2)
    qc.measure(2, 2)
    counts = sim.run(transpile(qc, sim), shots=shots).result().get_counts()
    bob_zero = sum(v for k, v in counts.items() if k[0] == "0")  # k[0] is c2
    return bob_zero / shots, counts

theta, phi = 1.1, 0.7                # an arbitrary "unknown" state
p_with, counts = teleport_test(theta, phi, True)
p_without, _ = teleport_test(theta, phi, False)

print("Counts (with corrections):", dict(sorted(counts.items())))
print(f"P(Bob reads 0) WITH corrections    : {p_with:.3f}   (expected 1.000)")
print(f"P(Bob reads 0) WITHOUT corrections : {p_without:.3f}   (expected about 0.5)")
```

Output of one real run

```
Counts (with corrections): {'000': 503, '001': 535, '010': 477, '011': 485}
P(Bob reads 0) WITH corrections    : 1.000   (expected 1.000)
P(Bob reads 0) WITHOUT corrections : 0.482   (expected about 0.5)
```

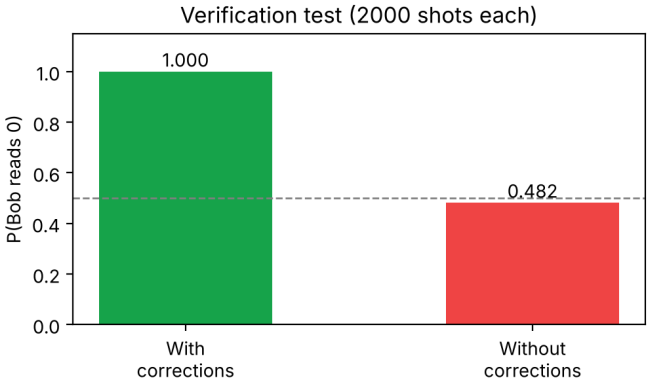


Figure 4: With the corrections, Bob reads 0 every time. Without them, the result is roughly random.

Reading the result. With corrections, every one of the 2000 shots has c2 = 0 (the keys start with 0), so Bob's qubit was exactly the prepared state. Alice's two bits are still uniformly random (four keys with about 500 counts each). Without corrections, the success rate collapses to about 50%, showing that the two classical bits are indispensable.

> **Important**
> The state used here, with θ = 1.1 rad and φ = 0.7 rad, is a stand-in for an “unknown” state: the teleportation circuit itself never uses θ or φ. Only the test does, to check the result.

11. Why Does This Not Violate Physics?

Question	Answer
Is information sent faster than light?	No. Before Bob receives Alice's two bits, his qubit is in a state that averages over the four cases; it is maximally mixed (completely random), whatever $\|\psi\rangle$ is. He can learn nothing about $\|\psi\rangle$ from it. Only after the bits arrive, at light speed or slower, can he fix his qubit.
Is no-cloning violated?	No. Alice's measurement destroys the original state. At no time do two copies exist.
Is energy or matter transported?	No. Bob's qubit was always at Bob's location.
Does Alice need to know $\|\psi\rangle$?	No. The protocol works for any unknown state, and Alice never learns α or β.
What is “instant”?	Bob's qubit changes its state conditional on Alice's result at the moment of her measurement. This is a correlation, and cannot be used to send a message without the classical bits.

Proof that Bob's qubit is random before the bits arrive. Averaging over the four equally likely cases gives $\frac{1}{4}[|\psi\rangle\langle\psi| + X|\psi\rangle\langle\psi|X + Z|\psi\rangle\langle\psi|Z + ZX|\psi\rangle\langle\psi|XZ] = I/2$, the maximally mixed state, for every $|\psi\rangle$. Bob's local description therefore does not depend on α and β at all.

12. History and Real Experiments

Year	Event
1993	Bennett, Brassard, Crépeau, Jozsa, Peres and Wootters propose quantum teleportation (*Teleporting an unknown quantum state via dual classical and Einstein-Podolsky-Rosen channels*, Physical Review Letters).
1997	First experimental teleportation of a photon's polarisation state, by Bouwmeester et al. in Innsbruck (Nature). A similar experiment in Rome (Boschi et al.) followed in 1998.
2000s–2010s	Teleportation demonstrated with atoms, ions, superconducting circuits and over progressively longer optical-fibre and free-space distances.
2017	Teleportation of a photon state from the ground to the Micius satellite over more than 1000 km (Ren et al., Nature).

Note: the dates and venues above are from my knowledge of the literature; if you cite them in a report, check the original papers.

13. Applications

Application	How teleportation is used
Quantum networks and quantum internet	Move qubits between nodes without sending the fragile qubit through a lossy channel
Quantum repeaters	Entanglement swapping (teleportation applied to half of an entangled pair) extends entanglement over long distances
Distributed quantum computing	Link small processors; “gate teleportation” performs a gate between distant qubits
Fault-tolerant quantum computing	Gate teleportation is a building block in some error-correction schemes
Quantum cryptography	Supports protocols that need shared entanglement

14. Troubleshooting

Problem	Cause	Fix
Error with if_test or c_if	Old or mismatched Qiskit/Aer	Upgrade; use if_test as in this guide
Only 4 keys show in counts	Mixed-up bit order, or too few shots	Print sorted keys; use 1000+ shots; remember the order c2 c1 c0
Results look biased	Wrong condition bit (swapped c0, c1)	X must depend on c1, Z must depend on c0
Verification gives P < 1	Missing correction, or inverse gates in wrong order	Undo RZ first, then RY (reverse of preparation)
Different counts each run	Normal: quantum measurement is probabilistic	Compare frequencies, not exact numbers
Real device cannot run if_test	Not all backends support dynamic circuits	Use a backend that supports them, or the deferred-measurement version (replace conditionals with qc.cx(1,2) and qc.cz(0,2))

14.1 Deferred-measurement version

Measurement-and-conditional-gate can be replaced by controlled gates before measurement. This is equivalent for the final state of q2 and avoids dynamic circuits:

```
# instead of the two if_test blocks, before any measurement:
qc.cx(1, 2)   # X on q2 controlled by q1
qc.cz(0, 2)   # Z on q2 controlled by q0
```

This is a useful trick for hardware that lacks mid-circuit measurement, but it is no longer a demonstration of classical communication, because no classical bits are sent.

15. Viva Questions and Answers

Q1. What is quantum teleportation?

A protocol that transfers an unknown quantum state from Alice's qubit to Bob's using a shared entangled pair and two classical bits.

Q2. Who proposed it and when?

Bennett, Brassard, Crépeau, Jozsa, Peres and Wootters, in 1993.

Q3. How many qubits does the circuit use?

Three: the message qubit, Alice's half of the pair and Bob's half.

Q4. How many classical bits are sent?

Two.

Q5. Does it transport matter?

No. It only re-creates the state on Bob's qubit.

Q6. Can it be used for faster-than-light communication?

No. Bob needs Alice's two classical bits, which cannot travel faster than light, and before that his qubit looks completely random.

Q7. What is entanglement?

A joint quantum state of several qubits that cannot be written as a product of individual states.

Q8. Which Bell state does the code create?

$|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$.

Q9. How do you create a Bell state?

Hadamard on one qubit, then CNOT with that qubit as control and the other as target.

Q10. What does CNOT do?

Flips the target qubit if the control is $|1\rangle$.

Q11. What does Alice's CNOT + H do?

Rotates the Bell basis into the computational basis, so that a normal measurement becomes a Bell measurement.

Q12. What are Alice's possible results and their probabilities?

(0,0), (0,1), (1,0), (1,1), each with probability 1/4, independent of the state being teleported.

Q13. What correction does Bob apply for each result?

(0,0): none; c1 = 1: X; c0 = 1: Z; both: X then Z.

Q14. Why does Bob need the classical bits?

Without them he cannot know which of four states he holds, so he cannot correct. His qubit is otherwise maximally mixed.

Q15. Is the original state preserved on Alice's side?

No. Her measurement destroys it, in agreement with no-cloning.

Q16. What is the no-cloning theorem?

An unknown quantum state cannot be copied exactly.

Q17. What is feed-forward?

Using a measurement result to control a later gate.

Q18. Why are the 8 outcomes equally likely in the code's output?

Alice's 2 bits are uniform (1/4 each) and Bob's final measurement of $|+\rangle$ is 50/50, independent, giving 1/8 each.

Q19. Does the basic histogram prove teleportation worked?

No, since $|+\rangle$ gives 50/50 in the Z basis anyway. Use the inverse-preparation test, where Bob should read 0 every time.

Q20. What is LOCC?

Local operations and classical communication, the allowed operations in teleportation (plus pre-shared entanglement).

Q21. What is entanglement swapping?

Applying teleportation to one half of an entangled pair, which entangles qubits that never interacted. It underlies quantum repeaters.

Q22. What is the correct order of the classical bits in Qiskit's output?

c2 c1 c0 from left to right: the right-most bit is classical bit 0.

Q23. What is the success probability of the protocol?

In the ideal case, 100%: it always works once Bob applies the right correction. On noisy hardware, fidelity is below 1.

16. Glossary

Term	Definition
Alice, Bob	Conventional names for the sender and receiver
Bell measurement	Joint measurement of two qubits in the Bell basis
Bell pair	A two-qubit entangled state such as $\|\Phi^+\rangle$
Classical channel	Ordinary communication link carrying bits
CNOT	Controlled-NOT two-qubit gate
Dynamic circuit	Circuit whose gates depend on mid-circuit measurement results
Entanglement	Non-product joint quantum state
Entanglement swapping	Teleportation applied to a half of an entangled pair
Feed-forward	Using measurement results to control later operations
Fidelity	Measure (0 to 1) of how close two quantum states are
Global phase	Overall factor $e^{iγ}$ on a state; not observable
Hadamard gate	Gate that maps $\|0\rangle$ to $(\|0\rangle+\|1\rangle)/\sqrt{2}$
Ket	Notation $\|ψ\rangle$ for a state vector
LOCC	Local operations and classical communication
Maximally mixed state	The state I/2: completely random, no information
No-cloning theorem	Unknown quantum states cannot be copied
Pauli gates (X, Y, Z)	Basic single-qubit gates; X flips bits, Z flips phase
Unitary	Reversible quantum operation ($U^\dagger U = I$)

17. Result and Conclusion

The quantum teleportation protocol was implemented using Python and Qiskit. The circuit entangled two qubits into a Bell pair, applied Alice's CNOT and Hadamard, measured her two qubits, and used the two classical results to apply X and Z corrections on Bob's qubit. The basic run produced eight equally likely outcomes, as predicted, and the inverse-preparation test showed that Bob's qubit held the prepared state in every one of 2000 shots with corrections, versus about 48% without them.

Key concept to remember: teleportation does not copy the state, move matter, or exceed light speed. Alice's measurement destroys her original; the shared entanglement plus two classical bits let Bob reconstruct the state on his qubit.

Five topics to focus on for your exam

1. Entanglement and the Bell state $|\Phi^+\rangle$, and how to create it
2. The algebra of Section 5 and the correction table
3. Why a CNOT + H followed by measurement is a Bell measurement
4. Classical feed-forward with if_test and the bit order c2 c1 c0
5. Why teleportation is not faster than light and is consistent with no-cloning